

Answer: D

- 6 When the car is traveling at maximum speed, there is no net force acting on the car. Hence, the driving force F_D on the car is cancelled out by the drag forces on the car D , i.e. $F_D = D$. Since $P = F_D v$, and $D = kv^2$ where k is the constant of proportionality,

$$\frac{P}{v} = kv^2$$

$$P = kv^3$$

$$\frac{P_1}{P_2} = \left(\frac{v_1}{v_2} \right)^3$$

$$\frac{90}{50} = \left(\frac{55}{v_2} \right)^3$$

$$v_2 = 45 \text{ m s}^{-1}$$

Answer: C

- 7 Work done by frictional force = change in gravitational potential energy + change in kinetic